Machine Learning: Support Vector Machine

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Content

- What is a Support Vector Machine(SVM)?
- Types of Support Vector Machine (SVM) Algorithms
- Important Terms
- How Does Support Vector Machine Work?
- Mathematical Intuition Behind Support Vector Machine
- Margin in Support Vector Machine
- Optimization Function and its Constraints
- Kernels in Support Vector Machine

What is a Support Vector Machine(SVM)?

Introduction

- SVM is a powerful supervised algorithm that works best on smaller datasets but on complex ones, is often implemented through an SVM model.

- Support Vector Machine, abbreviated as SVM can be used for both regression and classification tasks, but generally, they work best in classification problems.

- They were very famous around the time they were created, during the 1990s, and keep on being the go-to method for a high-performing algorithm with a little tuning.

What is a Support Vector Machine(SVM)?

Introduction

- Note: Don't get confused between SVM and logistic regression. Both the algorithms try to find the best hyperplane, but the main difference is logistic regression is a probabilistic approach whereas support vector machine is geometric, margin-based approach grounded in statistical learning theory.

- Now the question is which hyperplane does it select? There can be an infinite number of hyperplanes passing through a point and classifying the two classes perfectly. So, which one is the best?

- SVM does this by finding the maximum margin between the hyperplanes that means maximum distances between the two classes.

What is a Support Vector Machine(SVM)?

Logistic Regression vs Support Vector Machine (SVM)

- Depending on the number of features you have you can either choose Logistic Regression or SVM.

- SVM works best when the dataset is small and complex. It is usually advisable to first use logistic regression and see how does it performs, if it fails to give a good accuracy you can go for SVM without any kernel.

- Logistic regression and SVM without any kernel have similar performance but depending on your features, one may be more efficient than the other.

Content

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Types of Support Vector Machine (SVM) Algorithms

- Linear SVM: When the data is perfectly linearly separable only then we can use Linear SVM. Perfectly linearly separable means that the data points can be classified into 2 classes by using a single straight line(if 2D).

- Non-Linear SVM: When the data is not linearly separable then we can use Non-Linear SVM, which means when the data points cannot be separated into 2 classes by using a straight line (if 2D) then we use some advanced techniques like kernel tricks to classify them.

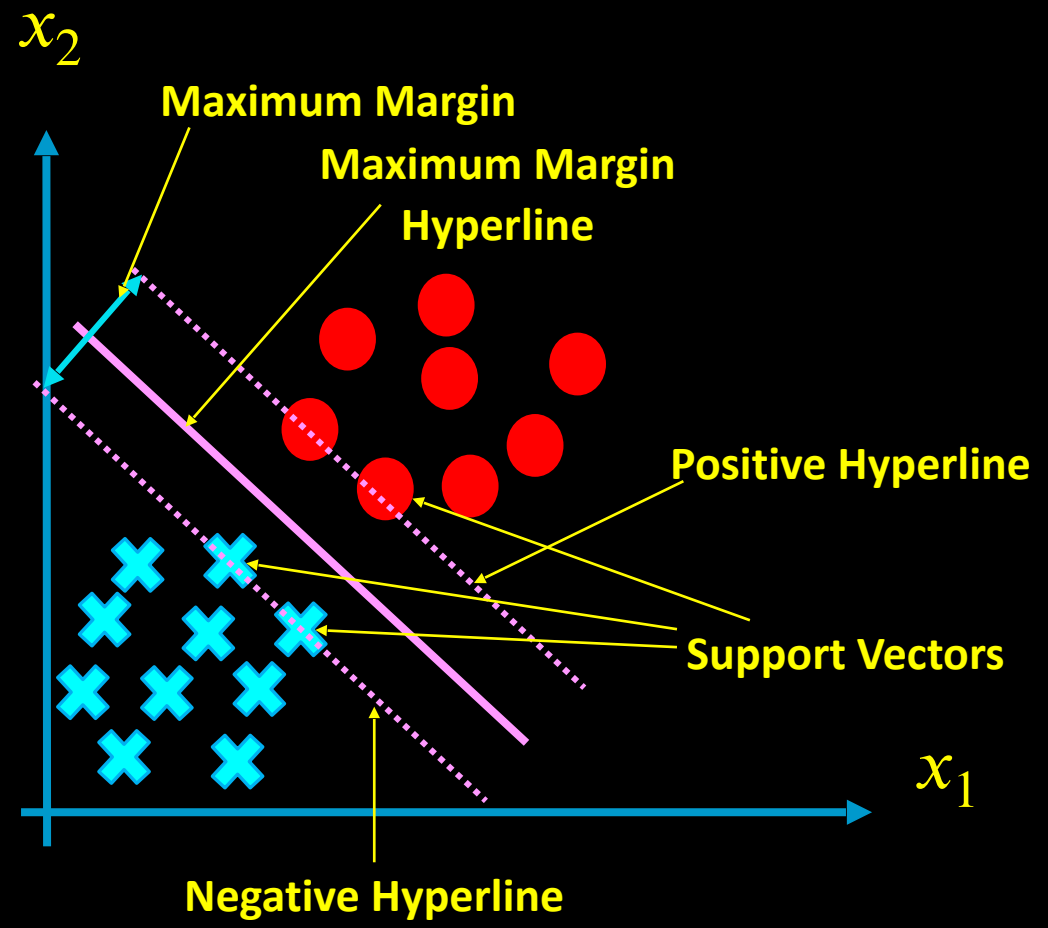
Content

- ~~What is a Support Vector Machine(SVM)?~~
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Important Terms

- Support Vectors: These are the points that are closest to the hyperplane. A separating line will be defined with the help of these data points.

- Margin: it is the distance between the hyperplane and the observations closest to the hyperplane (support vectors). In SVM large margin is considered a good margin. There are two types of margins hard margin and soft margin.



Content

- ~~What is a Support Vector Machine(SVM)?~~
- ~~Types of Support Vector Machine (SVM) Algorithms~~
- ~~Important Terms~~
- How Does Support Vector Machine Work?
- Mathematical Intuition Behind Support Vector Machine
- Margin in Support Vector Machine
- Optimization Function and its Constraints
- Kernels in Support Vector Machine

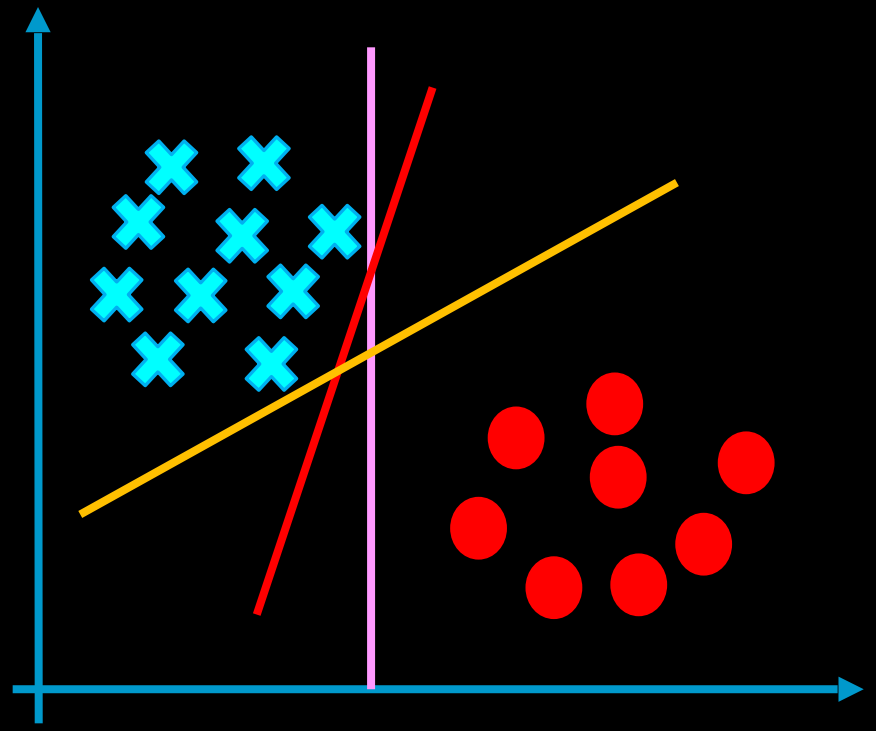
How Does SVM Work?

- SVM is defined such that it is defined in terms of the support vectors only, we don't have to worry about other observations since the margin is made using the points which are closest to the hyperplane (support vectors), whereas in logistic regression the classifier is defined over all the points.

How Does SVM Work?

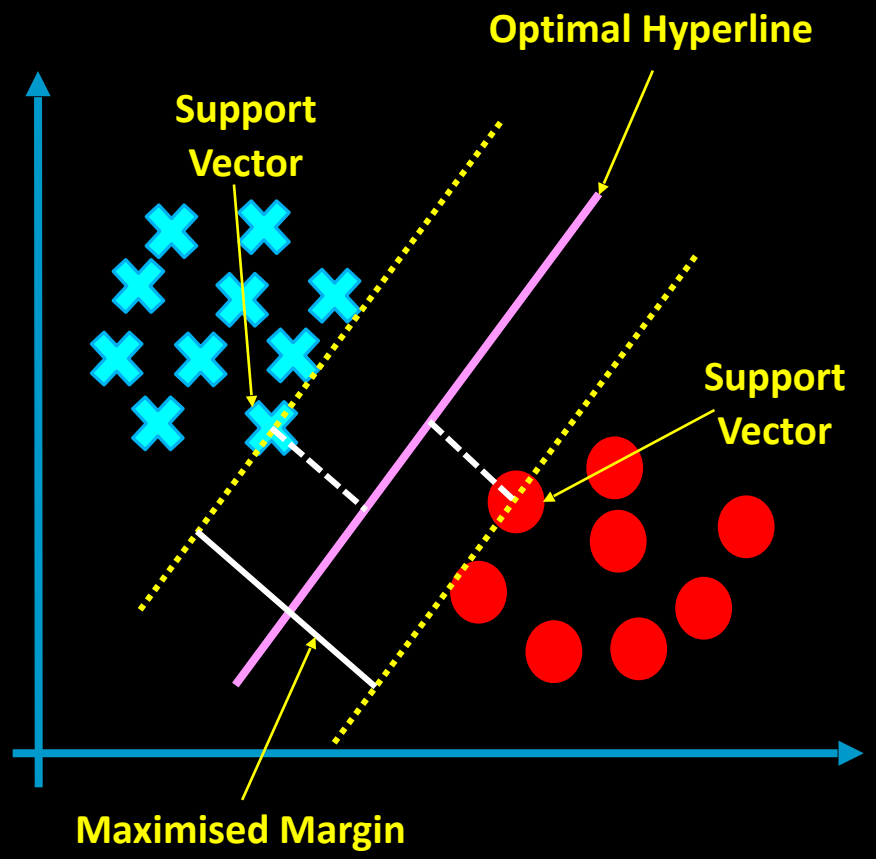
- To classify these points, we can have many decision boundaries, but the question is which is the best and how do we find it?

- NOTE: Since we are plotting the data points in a 2-dimensional graph we call this decision boundary a straight line but if we have more dimensions, we call this decision boundary a "hyperplane"



How Does SVM Work?

- The best hyperplane is that plane that has the maximum distance from both the classes, and this is the main aim of SVM.
- This is done by finding different hyperplanes which classify the labels in the best way then it will choose the one which is farthest from the data points or the one which has a maximum margin.



Content

- ~~What is a Support Vector Machine(SVM)?~~
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- ~~Important Terms~~
- ~~How Does Support Vector Machine Work?~~
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Mathematical Intuition Behind SVM

Understanding Dot-Product

- We all know that a vector is a quantity that has magnitude as well as direction and just like numbers we can use mathematical operations such as addition, multiplication.

- We will try to learn about the multiplication of vectors which can be done in two ways, dot product, and cross product.

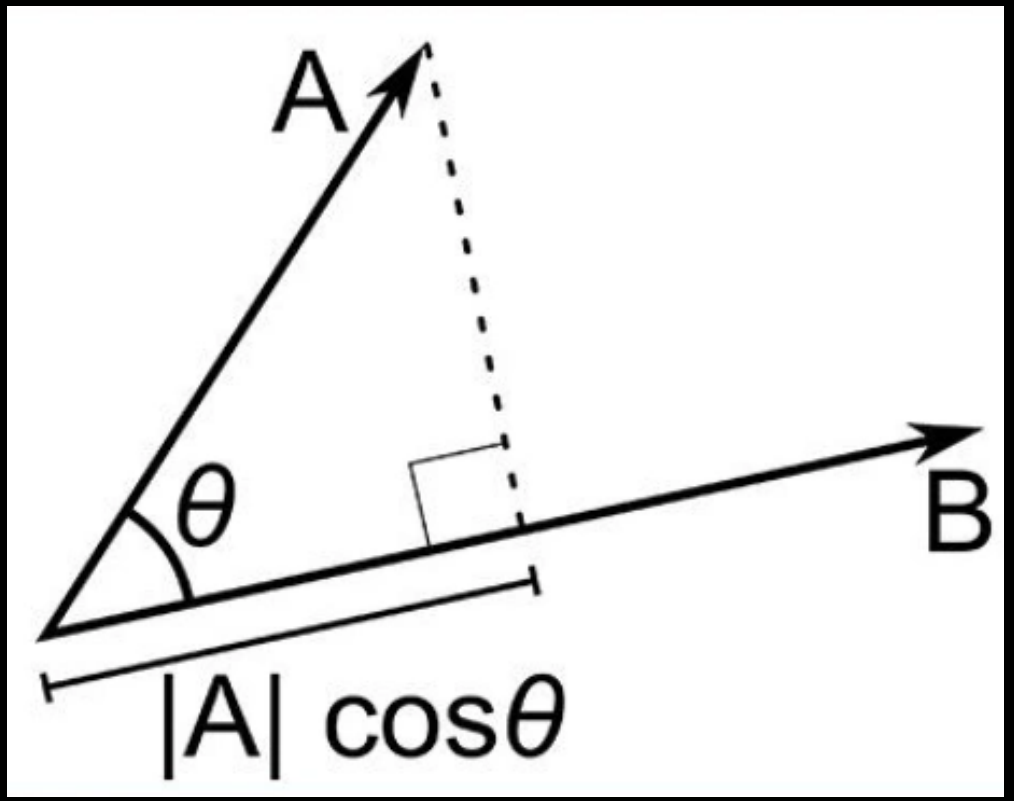
- The difference is only that the dot product is used to get a scalar value as a resultant whereas cross-product is used to obtain a vector again.

Mathematical Intuition Behind SVM

Understanding Dot-Product

- The dot product can be defined as the projection of one vector along with another, multiply by the product of another vector.

- Here A and B are 2 vectors, to find the dot product between these 2 vectors we first find the magnitude of both the vectors and to find magnitude we use the Pythagorean theorem or the distance formula.



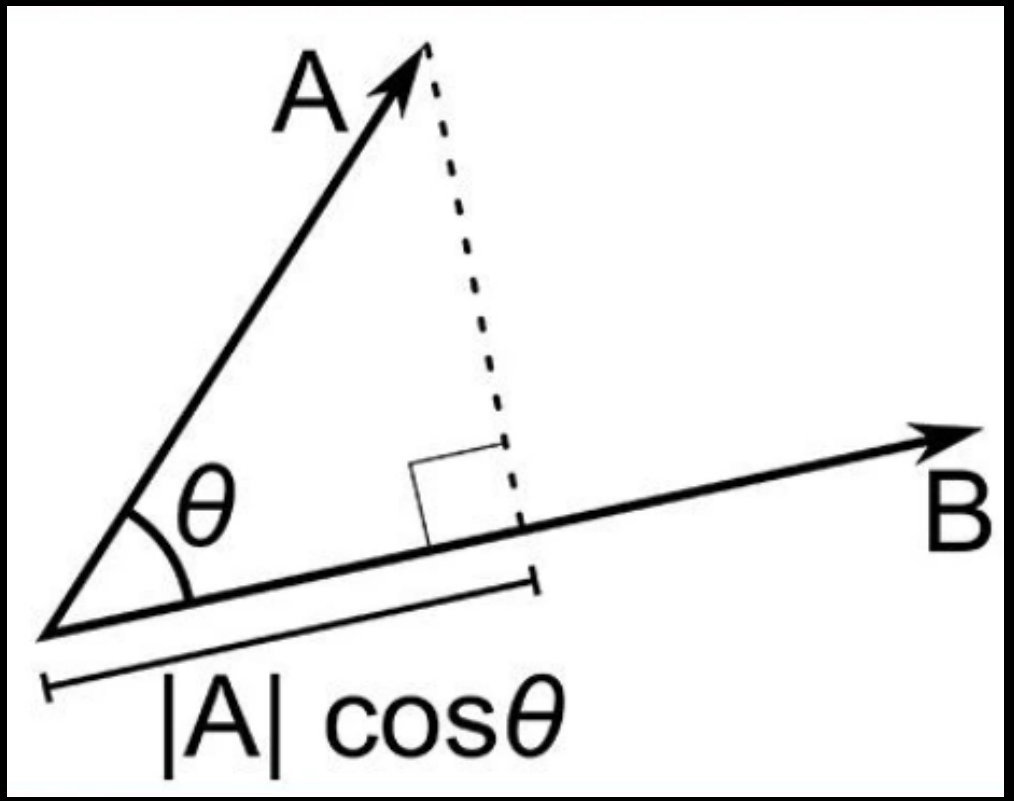
Mathematical Intuition Behind SVM

Understanding Dot-Product

➢ After finding the magnitude we simply multiply it with the cosine angle between both the vectors. Mathematically it can be written as:

$$A \,.\, B = |A| cos\theta * |B|$$

Where $|A| \cos\theta$ is the projection of A on B
And $|B|$ is the magnitude of vector B



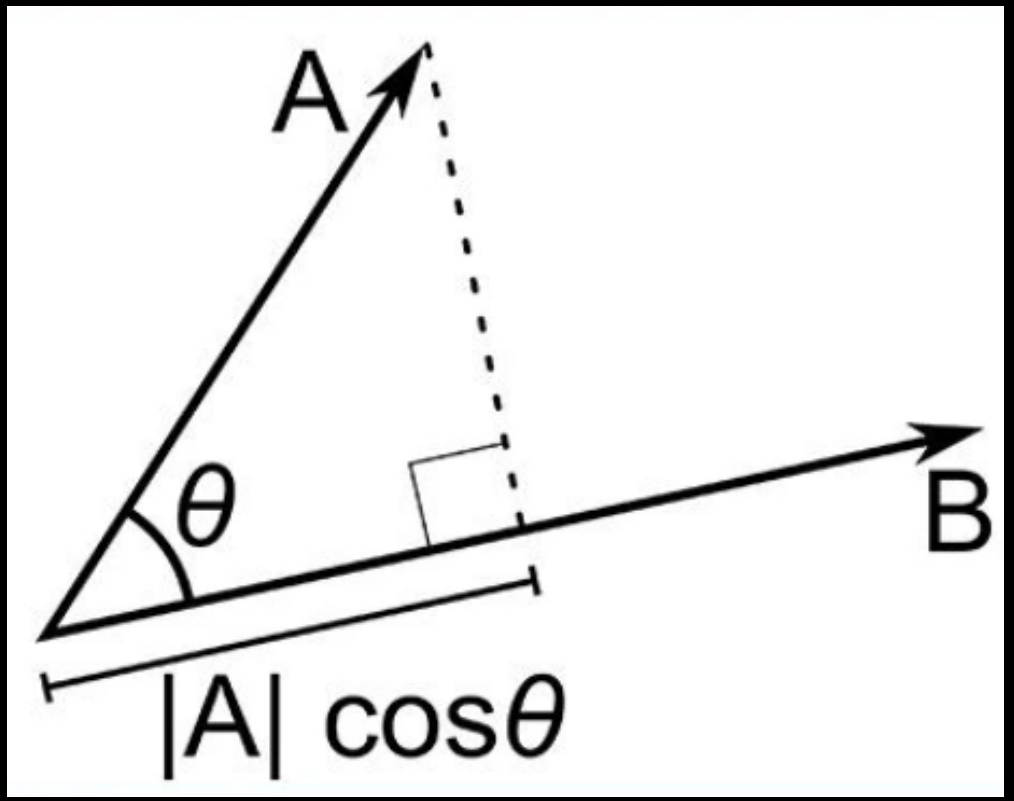
Mathematical Intuition Behind SVM

Understanding Dot-Product

Now in SVM we just need the projection of A not the magnitude of B, we will see why later.
To just get the projection we can simply take the unit vector of B because it will be in the direction of B but its magnitude will be 1.
Hence now the equation becomes:

$$A \,.\, B = |A| \cos\theta * unit\ vector\ of\ B$$

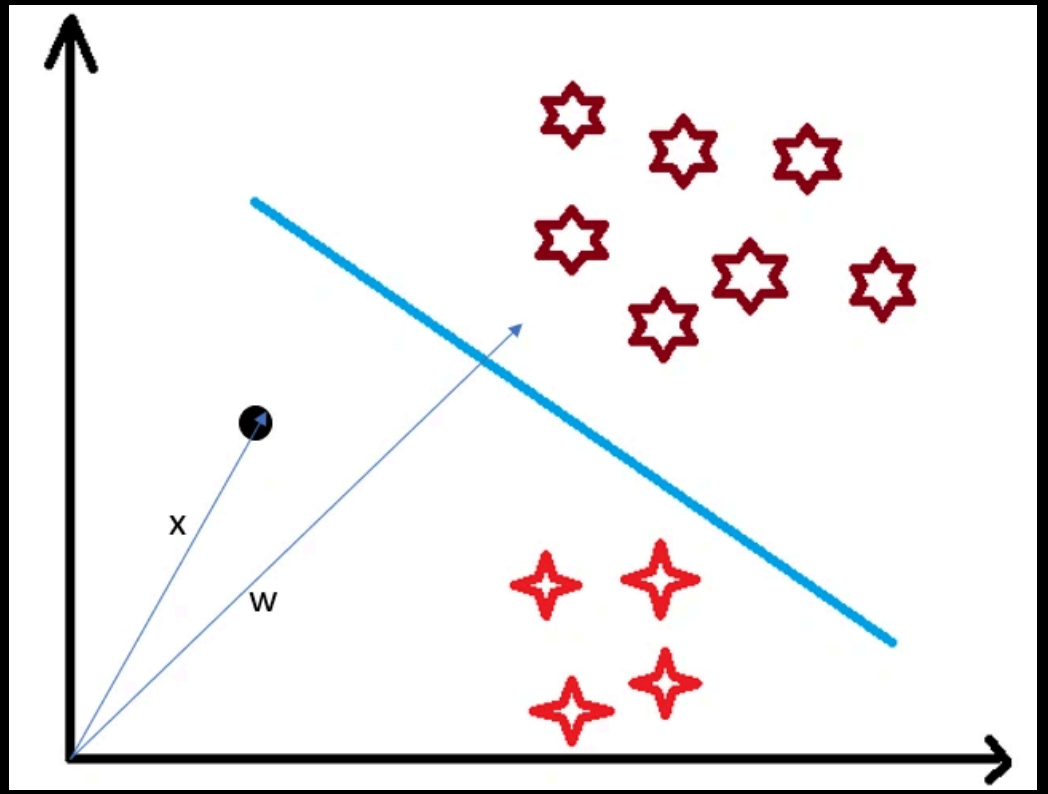
Now let's move to the next part and see how we will use this in SVM.



Mathematical Intuition Behind SVM

Use Dot-Product in SVM

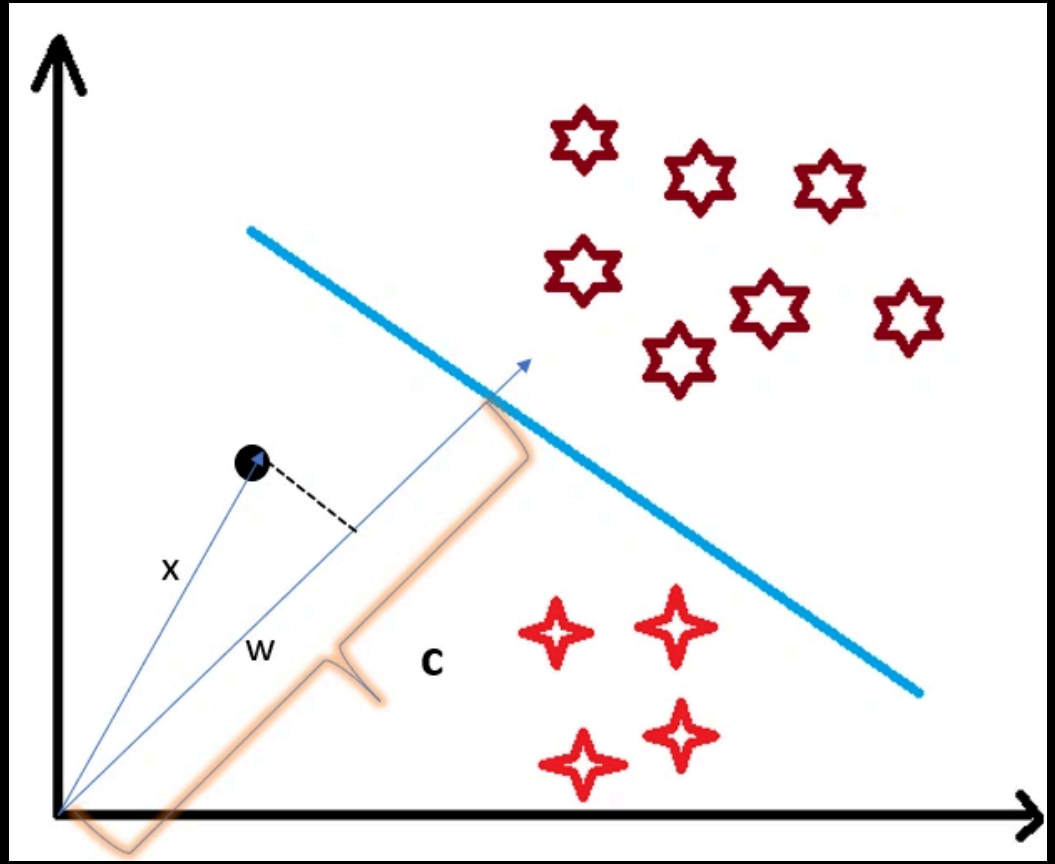
➢ Consider a random point X and we want to know whether it lies on the right side of the plane or the left side of the plane (positive or negative).



Mathematical Intuition Behind SVM

Use Dot-Product in SVM

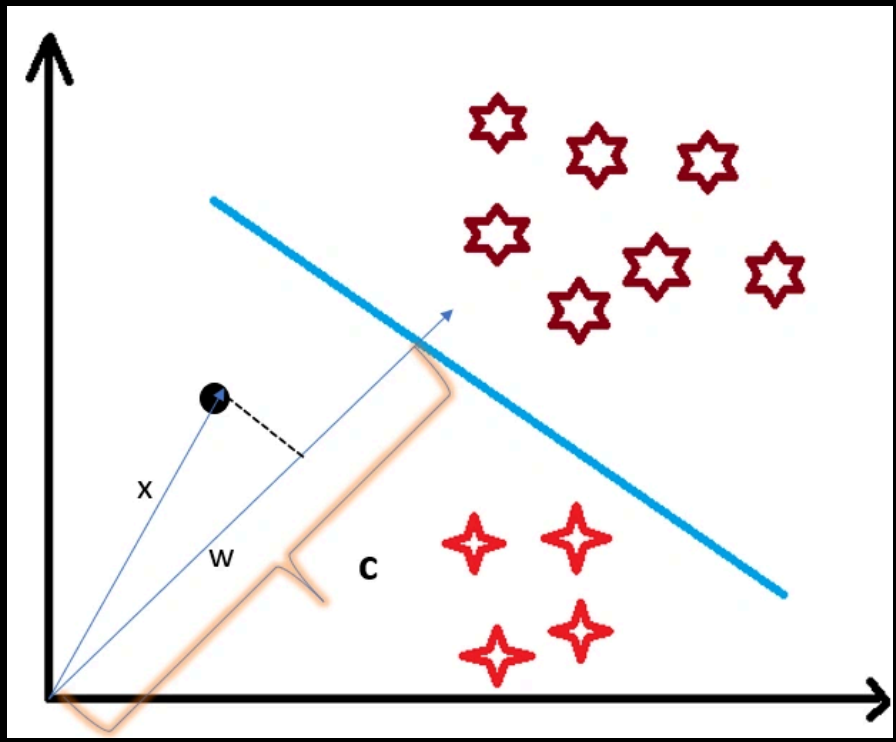
- To find this first we assume this point is a vector (X) and then we make a vector (w) which is perpendicular to the hyperplane.
- Let's say the distance of vector w from origin to decision boundary is 'c'.
- Now we take the projection of X vector on w.



Mathematical Intuition Behind SVM

Use Dot-Product in SVM

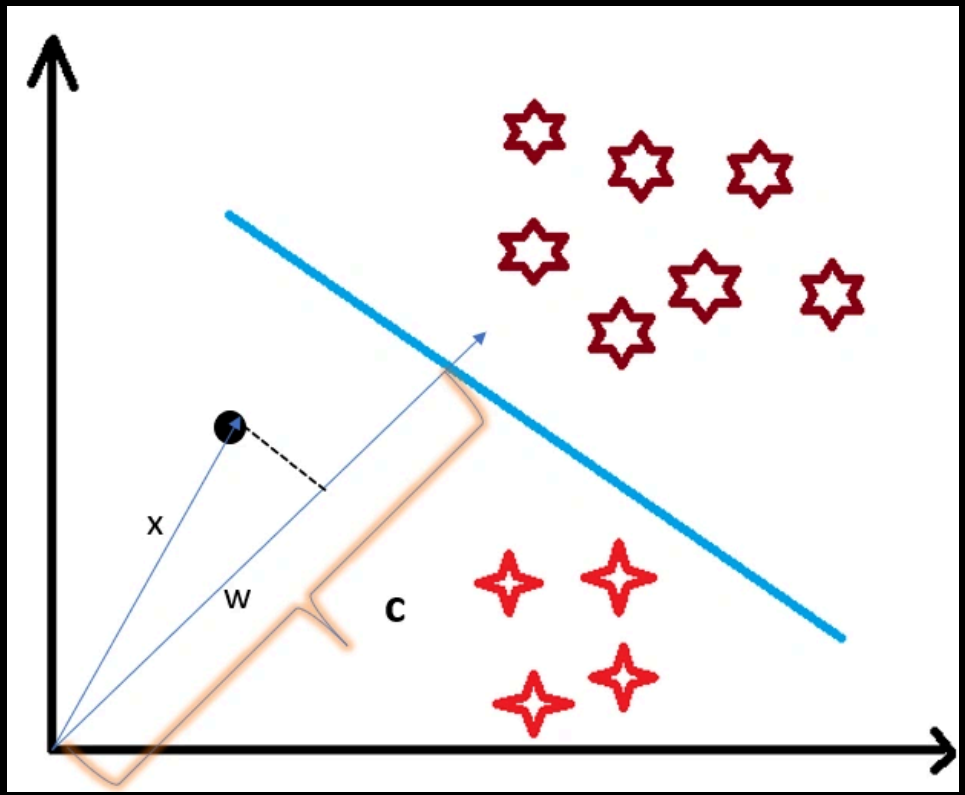
- $\vec{X} \cdot w = c$ (*the point lies on the decision boundary*)
- $\vec{X} \cdot w > c$ (*Positive samples*)
- $\vec{X} \cdot w < c$ (*Negative samples*)



Mathematical Intuition Behind SVM

Use Dot-Product in SVM

- You must be having this doubt that why did we take this perpendicular vector w to the hyperplane?
- So what we want is the distance of vector X from the decision boundary and there can be infinite points on the boundary to measure the distance from.
- So that's why we come to standard, we simply take perpendicular and use it as a reference and then take projections of all the other data points on this perpendicular vector and then compare the distance.

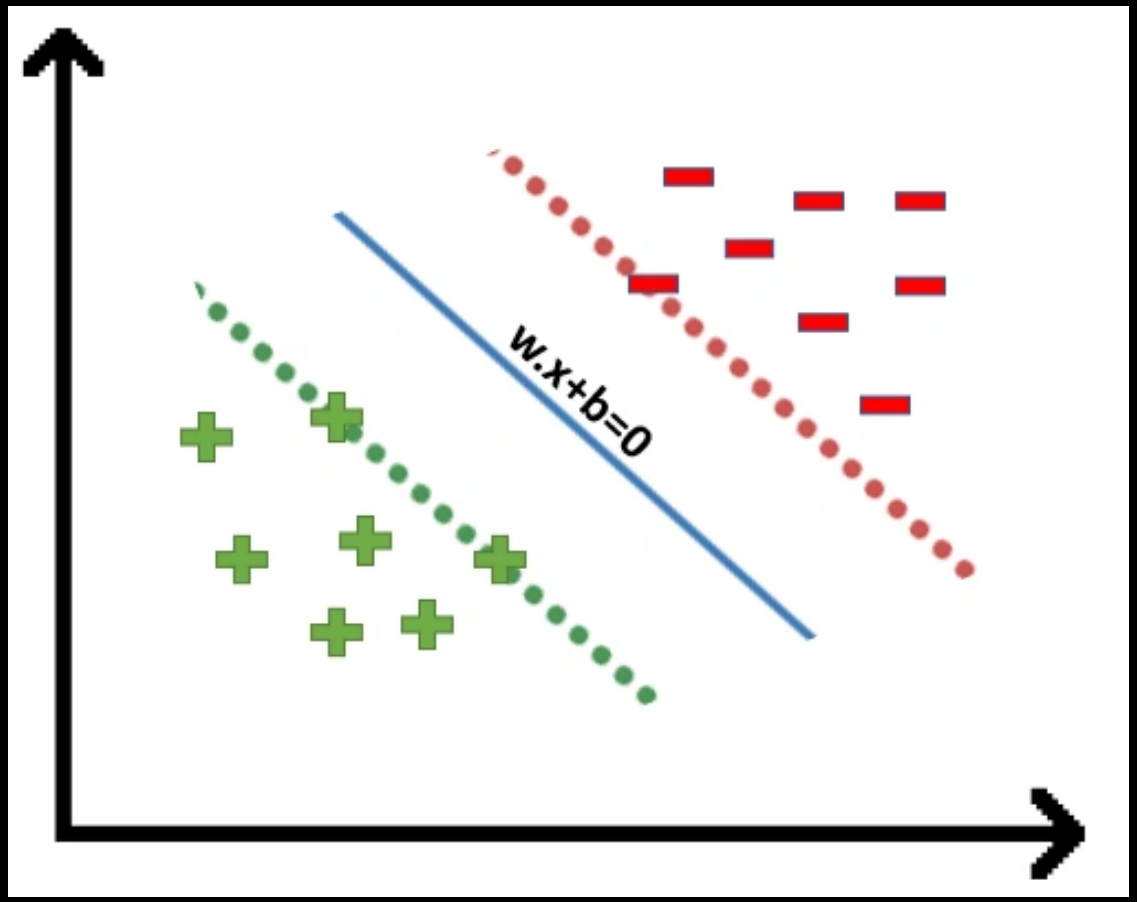


Content

- ~~What is a Support Vector Machine(SVM)?~~
- ~~Types of Support Vector Machine (SVM) Algorithms~~
- ~~Important Terms~~
- ~~How Does Support Vector Machine Work?~~
- ~~Mathematical Intuition Behind Support Vector Machine~~
- Margin in Support Vector Machine
- Optimization Function and its Constraints
- Kernels in Support Vector Machine

Margin in Support Vector Machine

➢ We all know the equation of a hyperplane is $\boldsymbol{w} \cdot \boldsymbol{x} + \boldsymbol{b} = \mathbf{0}$ where w is a vector normal to hyperplane and b is an offset.



Margin in Support Vector Machine

- To classify a point as negative or positive we need to define a decision rule.
- We can define decision rule as:

$$\vec{X}.\vec{w} - \mathrm{c} \geq 0$$

putting $-\mathrm{c}$ as b, we get

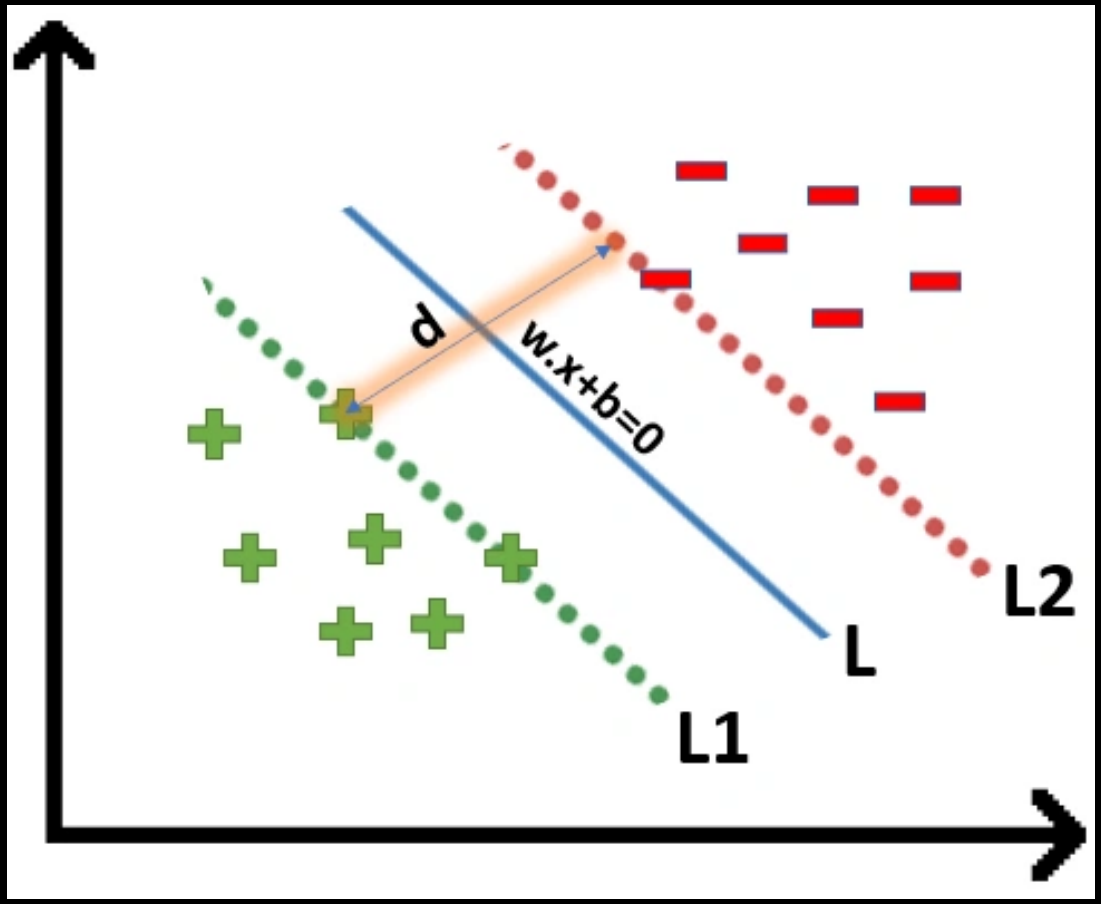
$$\vec{X}.\vec{w} + \mathrm{b} \geq 0$$

hence

$$\mathrm{y} = \begin{cases} +1 & \text{if } \vec{X}.\vec{w} + \mathrm{b} \geq 0 \\ -1 & \text{if } \vec{X}.\vec{w} + \mathrm{b} < 0 \end{cases}$$

Margin in Support Vector Machine

- If the value of $w \,.\, x + b \geq 0$ then we can say it is a positive point otherwise it is a negative point.
- Now we need (w, b) such that the margin has a maximum distance. Let's say this distance is 'd'.
- To calculate 'd' we need the equation of L1 and L2.
- For this, we will take few assumptions that the equation of L1 is $\boldsymbol{w \,.\, x + b = 1}$ and for L2 it is $\boldsymbol{w \,.\, x + b = -1}$.



Content

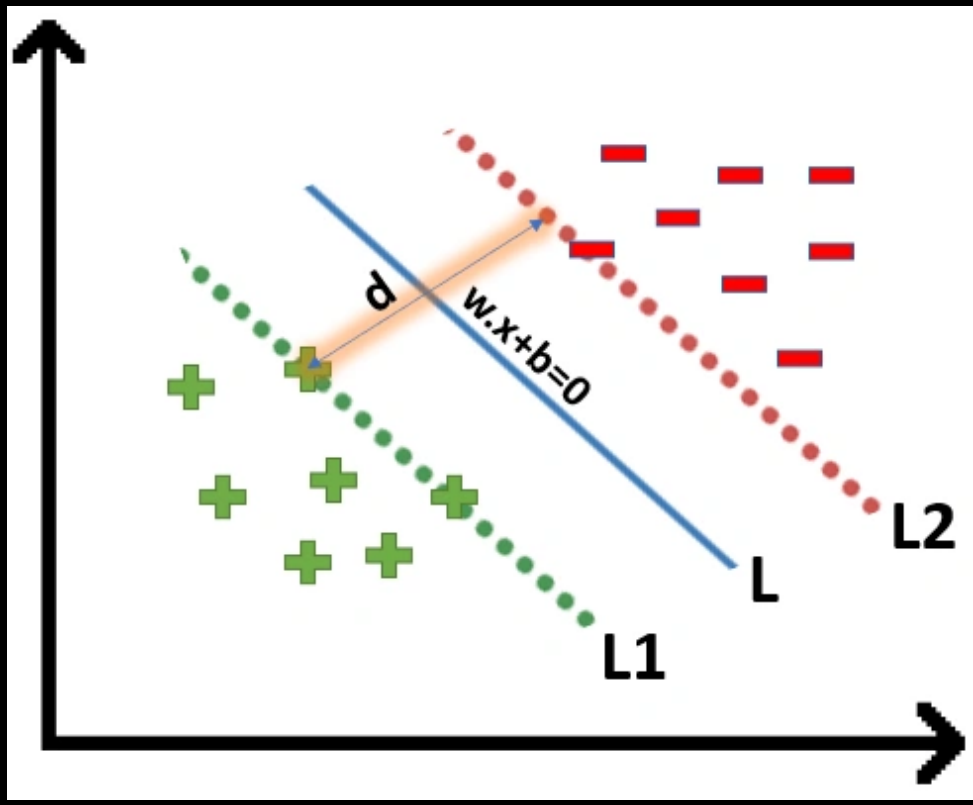
- ~~What is a Support Vector Machine(SVM)?~~
- ~~Types of Support Vector Machine (SVM) Algorithms~~
- ~~Important Terms~~
- ~~How Does Support Vector Machine Work?~~
- ~~Mathematical Intuition Behind Support Vector Machine~~
- ~~Margin in Support Vector Machine~~
- Optimization Function and its Constraints
- Kernels in Support Vector Machine

Optimization Function and its Constraints

- In order to get our optimization function, there are few constraints to consider. That constraint is that "We'll calculate the distance (d) in such a way that no positive or negative point can cross the margin line".

- Let's write these constraints mathematically:

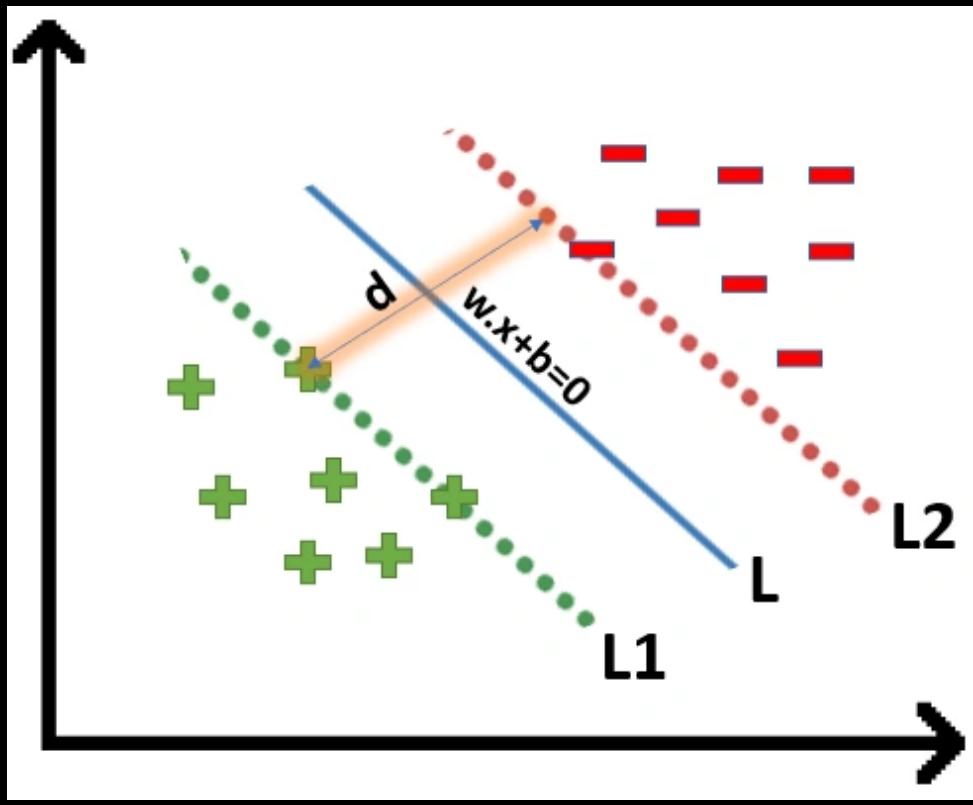
$$For\ all\ the\ Red\ point\ \vec{w} \cdot \vec{X} + b \leq -1$$
$$For\ all\ the\ Green\ point\ \vec{w} \cdot \vec{X} + b \geq 1$$



Optimization Function and its Constraints

- Rather than taking 2 constraints forward, we'll now try to simplify these two constraints into 1. We assume that negative classes have $y = -1$ and positive classes have $y = 1$.
- We can say that for every point to be correctly classified this condition should always be true:

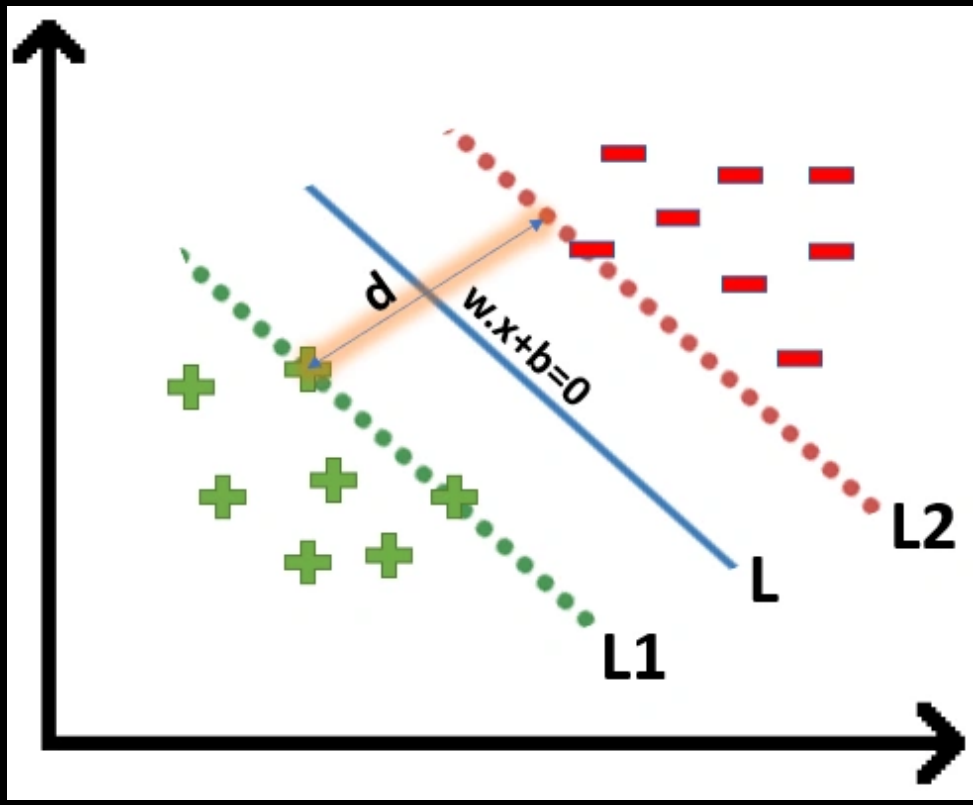
$$y_i(\vec{w} \,.\, \vec{X} + b) \geq 1$$



Optimization Function and its Constraints

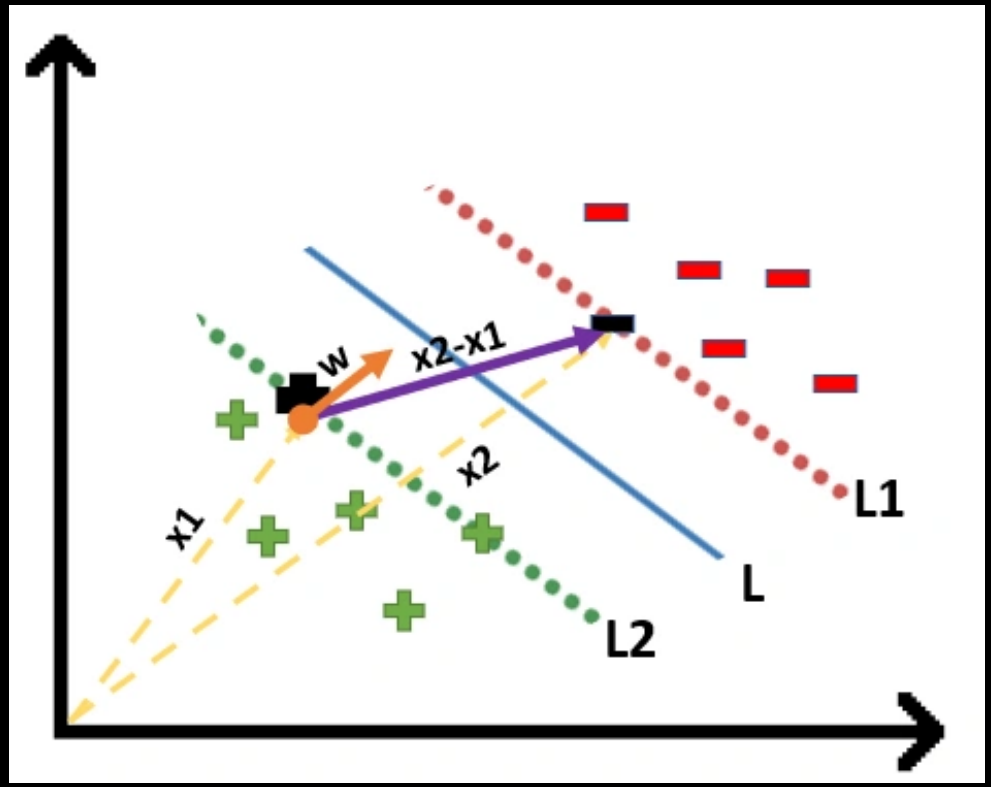
- Suppose a green point is correctly classified that means it will follow $w \cdot x + b >= 1$, if we multiply this with $y = 1$ we get this same equation mentioned above.

- Similarly, if we do this with a red point with $y = -1$ we will again get this equation. Hence, we can say that we need to maximize (d) such that this constraint holds true.



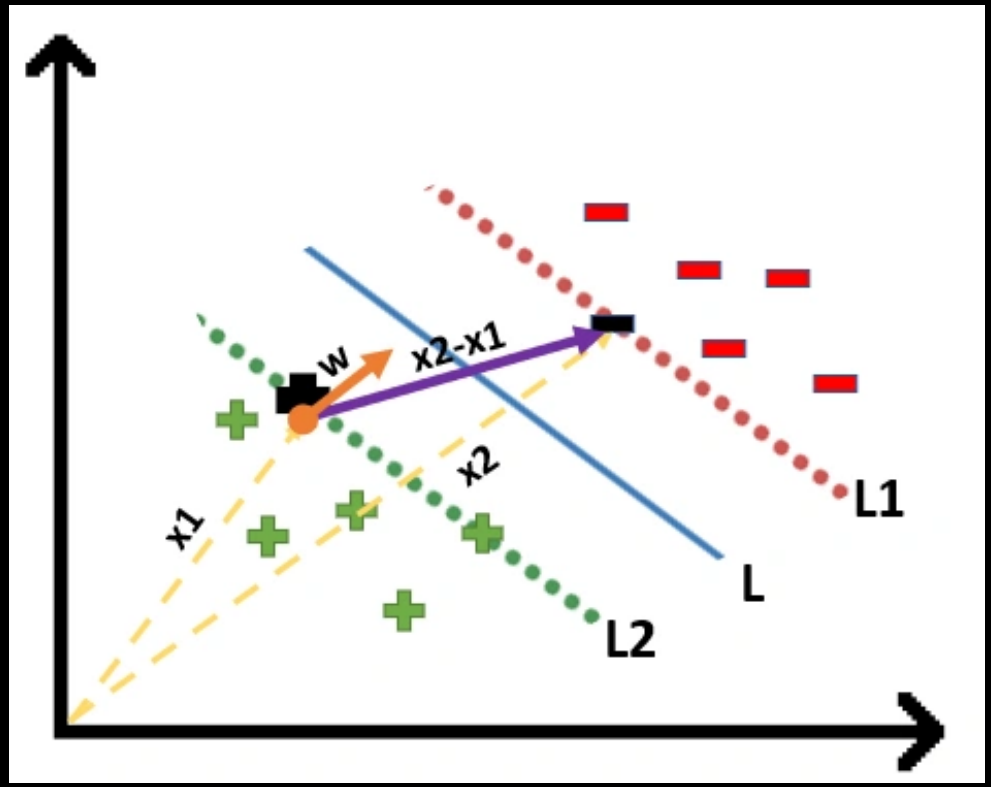
Optimization Function and its Constraints

- We will take 2 support vectors, 1 from the negative class and 2nd from the positive class.
- The distance between these two vectors $x1$ and $x2$ will be $(x2 - x1)$ vector.
- What we need is, the shortest distance between these two points which can be found using a trick we used in the dot product.
- We take a vector 'w' perpendicular to the hyperplane and then find the projection of $(x2 - x1)$ vector on 'w'.



Optimization Function and its Constraints

- Note: this perpendicular vector should be a unit vector then only this will work. Why this should be a unit vector? This has been explained in the dot-product section. To make this ‘w’ a unit vector we divide this with the norm of ‘w’.



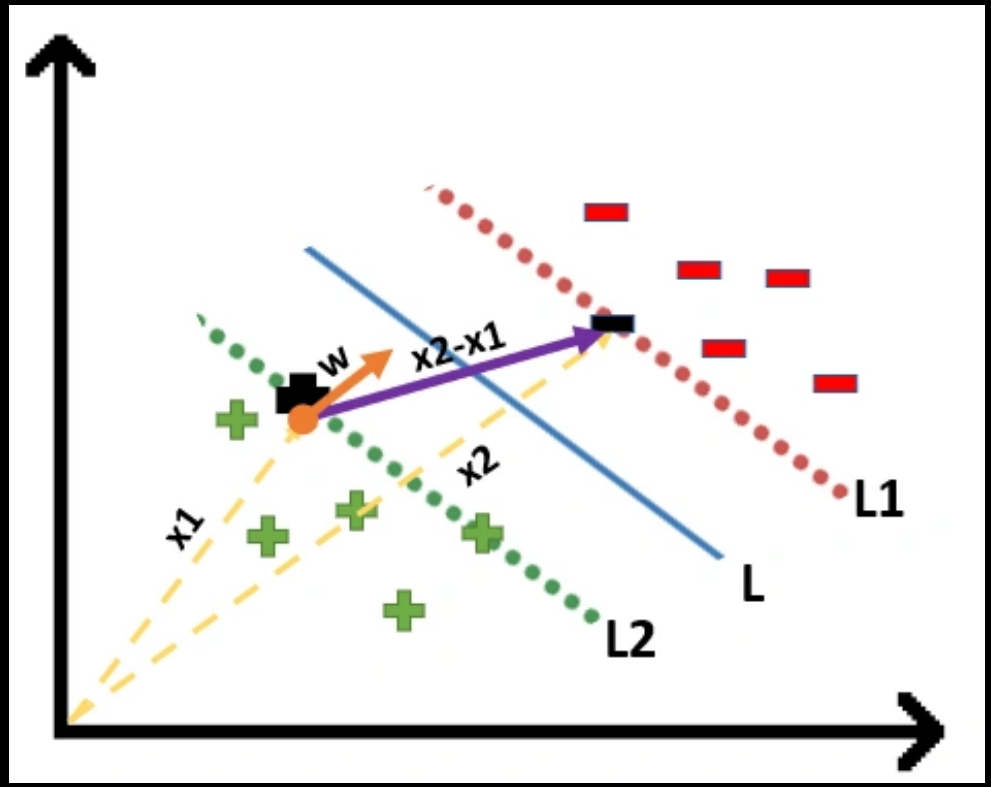
Optimization Function and its Constraints

Finding Projection of a Vector on Another Vector Using Dot Product

- We already know how to find the projection of a vector on another vector. We do this by dot-product of both vectors.
- So let's see how:

$$\Rightarrow \quad (x2 - x1) . \frac{\vec{w}}{\|w\|}$$

$$\Rightarrow \quad \frac{x2 . \vec{w} - x1 . \vec{w}}{\|w\|} \qquad ----(1)$$



Optimization Function and its Constraints

Finding Projection of a Vector on Another Vector Using Dot Product

➢ Since $x2$ and $x1$ are support vectors and they lie on the hyperplane, hence they will follow $yi * (2.x + b) = 1$ so we can write it as:

for positive point $y = 1$

$$\Rightarrow 1 \times (\vec{w}.x1 + b) = 1$$

$$\Rightarrow \vec{w}.x1 = 1 - b \quad -----(2)$$

Similarly for negative point $y = -1$

$$\Rightarrow -1 \times \left(\vec{w}.x2 + b\right) = 1$$

$$\Rightarrow \vec{w}.x2 = -b - 1 \quad -----(3)$$

Optimization Function and its Constraints

Finding Projection of a Vector on Another Vector Using Dot Product

➢ Putting equations (2) and (3) in equation (1) we get:

$$\Rightarrow \frac{(1-b)-(-b-1)}{\|w\|}$$

$$\Rightarrow \frac{1-b+b+1}{\|w\|} = \frac{2}{\|w\|} = d$$

Optimization Function and its Constraints

Finding Projection of a Vector on Another Vector Using Dot Product

➢ Hence the equation which we have to maximize is:

$$\text{argmax}(\text{w}^*, \text{b}^*) \ \frac{2}{\|\text{w}\|} \text{ such that } \text{y}_\text{i}\left(\vec{w}.\vec{X} + b\right) \geq 1$$

$$minimize,\ J(w,b) = \frac{1}{2}||w||^2$$

Optimization Function and its Constraints

Finding Projection of a Vector on Another Vector Using Dot Product

- We have now found our optimization function but there is a catch here that we don't find this type of perfectly linearly separable data in the industry, there is hardly any case we get this type of data and hence we fail to use this condition we proved here.

- The type of problem which we just studied is called Hard Margin SVM now we shall study soft margin which is similar to this but there are few more interesting tricks we use in Soft Margin SVM.

Optimization Function and its Constraints

Soft Margin SVM

- In real-life applications, we rarely encounter datasets that are perfectly linearly separable.
- Instead, we often come across datasets that are either nearly linearly separable or entirely non-linearly separable.
- Unfortunately, the trick demonstrated above for linearly separable datasets is not applicable in these cases.

- This is where SVM come into play. These are a powerful tool in machine learning that can effectively handle both almost linearly separable and non-linearly separable datasets, providing a robust solution to classification problems in diverse real-world scenarios.

Optimization Function and its Constraints

Soft Margin SVM

➢ To tackle this problem what we do is modify that equation in such a way that it allows few misclassifications that means it allows few points to be wrongly classified.

➢ We know that $\max[f(x)]$ can also be written as $\min[1/f(x)]$, it is common practice to minimize a cost function for optimization problems; therefore, we can invert the function.

$$\text{argmin}(\mathrm{w}^*, \mathrm{b}^*) \frac{\|\mathrm{w}\|}{2} \text{ such that } \mathrm{y_i}\left(\vec{w}.\vec{X} + b\right) \geq 1$$

Optimization Function and its Constraints

Soft Margin SVM

➢ To make a soft margin equation we add 2 more terms to this equation which is zeta and multiply that by a hyperparameter ‘c’

$$\operatorname{argmin}\left(\mathrm{w}^{*}, \mathrm{b}^{*}\right) \frac{\|\mathrm{w}\|}{2} + c\sum_{i=1}^{n}\zeta_i$$

$$J(w,b) = \frac{1}{2}||w||^2 + C\sum_{i=1}^{N}\max(0, 1 - y_i(w^T x_i + b))$$

Optimization Function and its Constraints

Soft Margin SVM

➢ To make a soft margin equation we add 2 more terms to this equation which is zeta and multiply that by a hyperparameter ‘c’

$$J(w, b) = \frac{1}{2}||w||^2 + C\sum_{i=1}^{N} \max(0, 1 - y_i(w^T x_i + b))$$

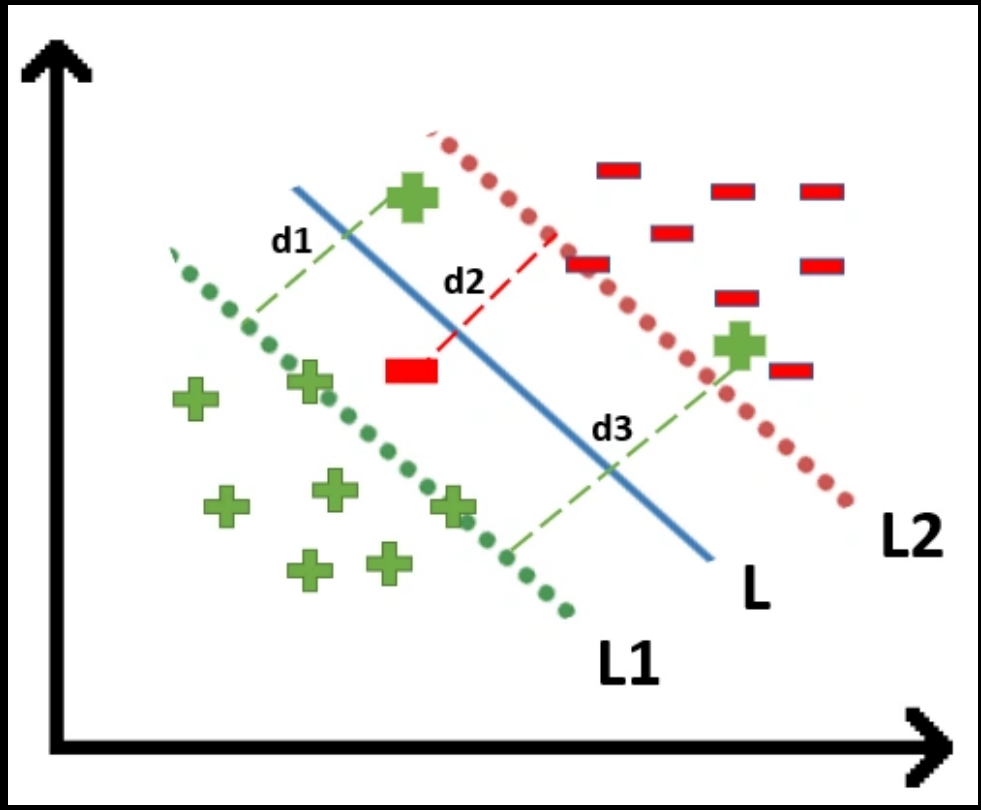
C control the same trade-off (margin vs. errors), but in opposite ways:

- Large C → model penalizes violations strongly → smaller margin, fewer misclassifications.
- Small C → allows more violations → wider margin.

Optimization Function and its Constraints

Soft Margin SVM

- For all the correctly classified points our zeta will be equal to 0 and for all the incorrectly classified points the zeta is simply the distance of that particular point from its correct hyperplane that means if we see the wrongly classified green points the value of zeta will be the distance of these points from L1 hyperplane and for wrongly classified redpoint zeta will be the distance of that point from L2 hyperplane.

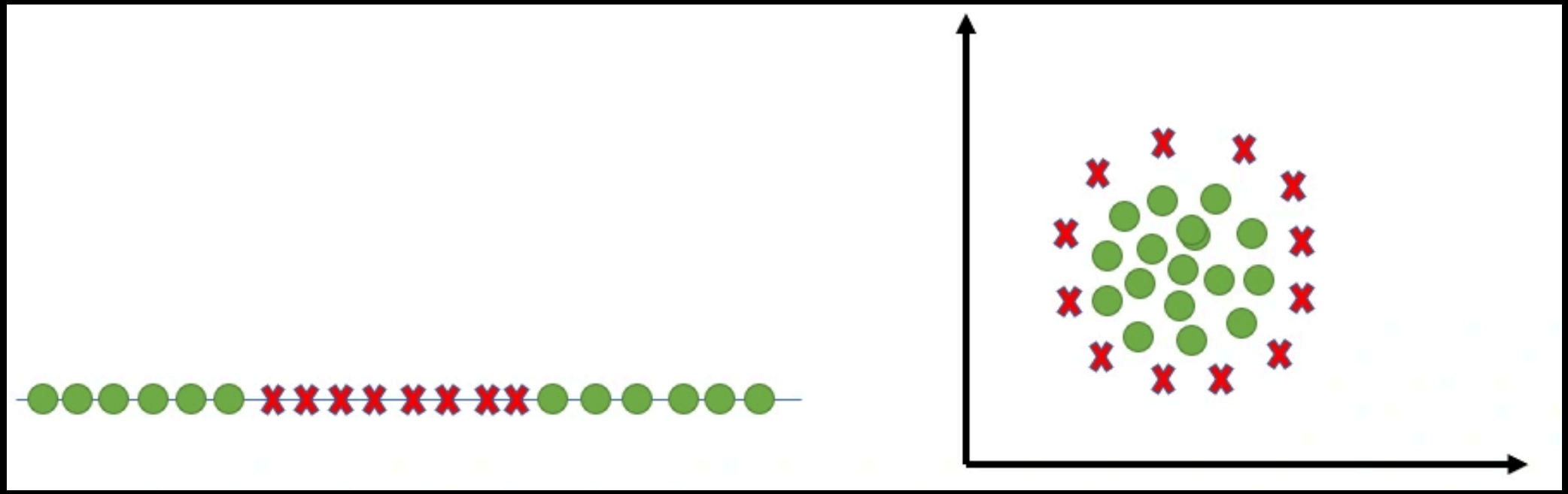


Content

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Kernels in Support Vector Machine

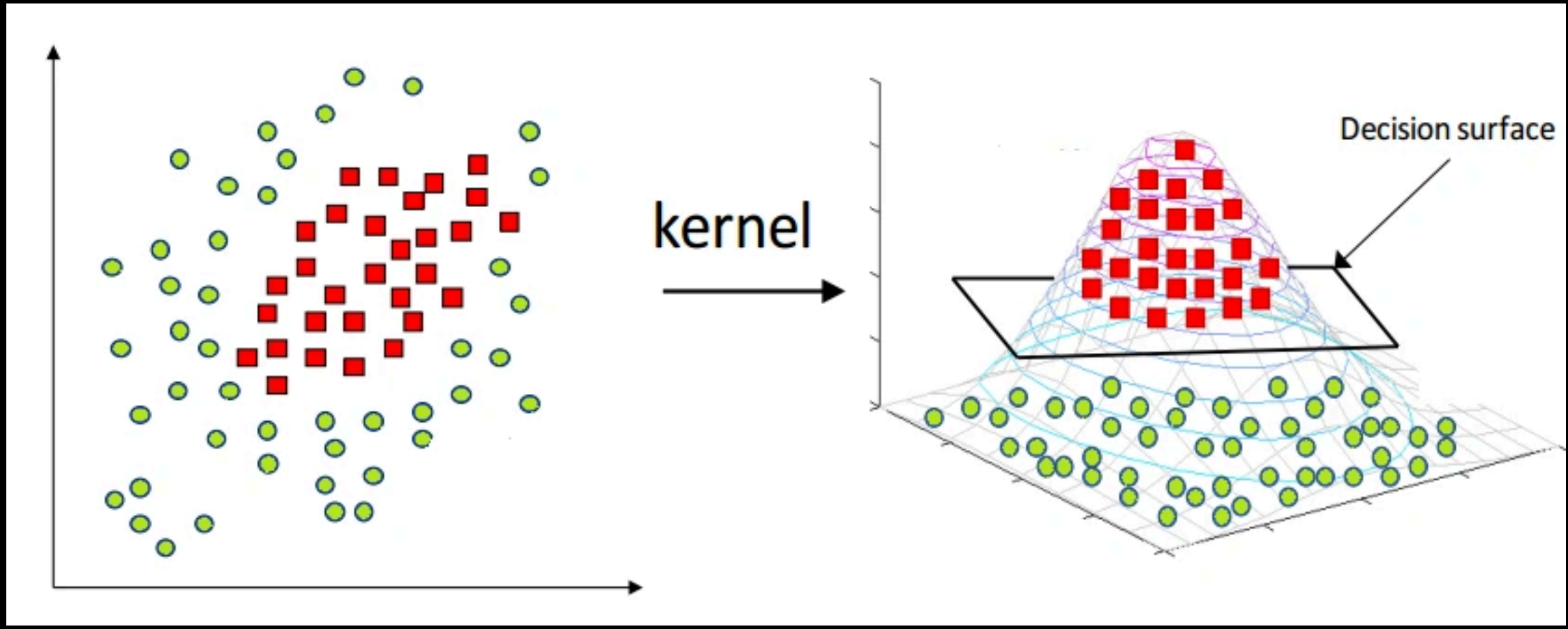
➢ The most interesting feature of SVM is that it can even work with a non-linear dataset and for this, we use "Kernel Trick" which makes it easier to classifies the points. Suppose we have a dataset like this:



Kernels in Support Vector Machine

- Here we see we cannot draw a single line or say hyperplane which can classify the points correctly.
- So what we do is try converting this lower dimension space to a higher dimension space using some quadratic functions which will allow us to find a decision boundary that clearly divides the data points.
- These functions which help us do this are called Kernels and which kernel to use is purely determined by hyperparameter tuning.

Kernels in Support Vector Machine



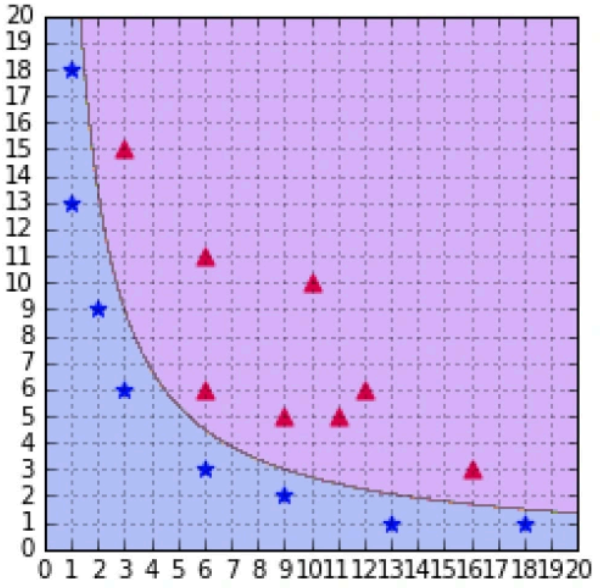
Kernels in Support Vector Machine

Polynomial Kernel

Following is the formula for the polynomial kernel:

$$f(X1, X2) = (X1^T . X2 + 1)^d$$

Here d is the degree of the polynomial, which we need to specify manually.



A SVM using a polynomial kernel is able to separate the data (degree=2)

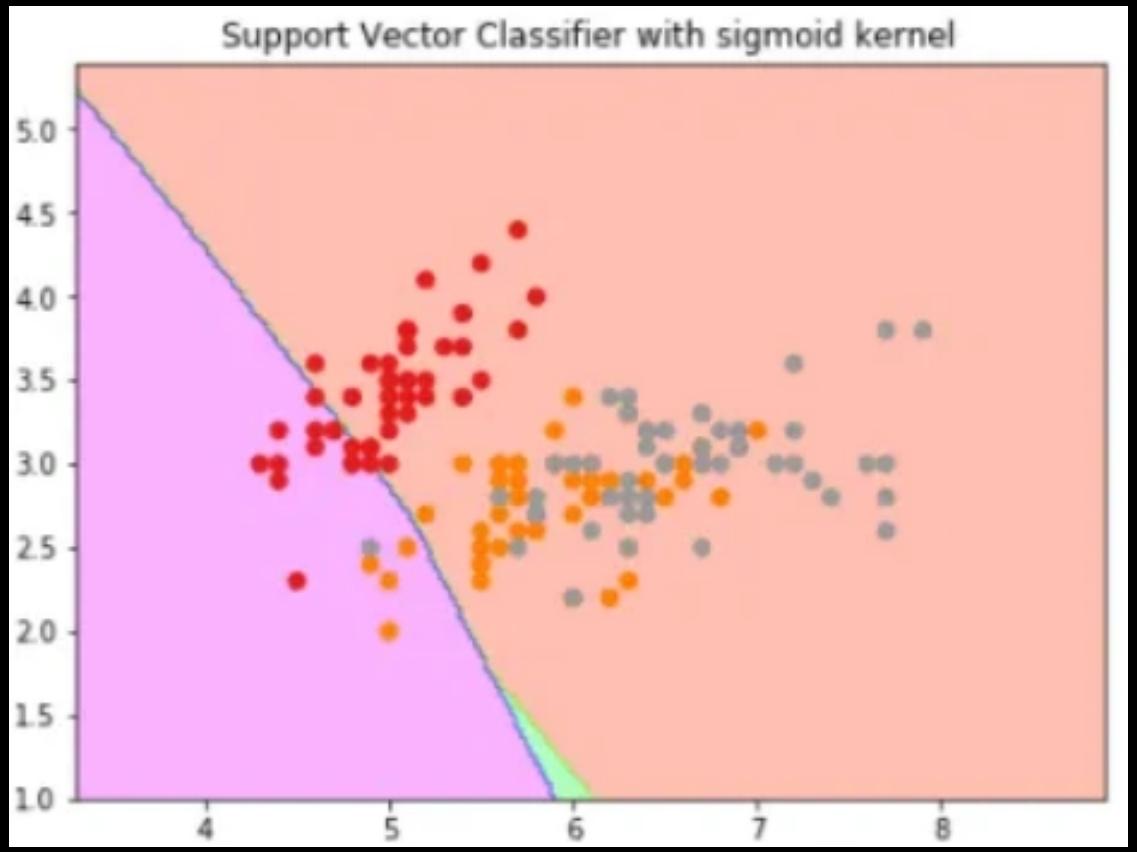
Kernels in Support Vector Machine

Sigmoid Kernel

We can use it as the proxy for neural networks. Equation is:

$$f(x1, x2) = \tanh(\alpha x^T y + x)$$

It is just taking your input, mapping them to a value of 0 and 1 so that they can be separated by a simple straight line.



Kernels in Support Vector Machine

Advantages of SVM

- SVM works better when the data is Linear
- It is more effective in high dimensions
- With the help of the kernel trick, we can solve any complex problem
- SVM is not sensitive to outliers
- Can help us with Image classification

Disadvantages of SVM

- Choosing a good kernel is not easy
- It doesn't show good results on a big dataset
- The SVM hyperparameters are Cost -C and gamma. It is not that easy to fine-tune these hyper-parameters. It is hard to visualize their impact




THANK YOU